

Who Survived the Titanic?

An Exploratory Analysis of Survival Patterns During the 1912 Maritime Disaster

Course Project: R for Data Science

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Executive Summary

This report explores the built-in `Titanic` dataset from the R `datasets` package. The dataset captures survival outcomes from the Titanic disaster in a compact contingency table format representing 2,201 passengers and crew members.

The central question is simple:

Who was more likely to survive the Titanic disaster, and which passenger characteristics mattered most?

The analysis shows how survival varied by passenger class, sex, and age. It also uses chi-square tests and logistic regression on the aggregated passenger counts to quantify the relationship between survival and passenger profile.

To extend the analysis beyond descriptive statistics, a Logistic Regression Machine Learning model was developed to predict passenger survival using passenger class, sex, and age category as predictor variables. The model was evaluated using a train-test split approach, confusion matrix, accuracy metrics, sensitivity, and specificity measures. Results indicate that survival outcomes can be predicted with meaningful accuracy, highlighting the importance of demographic and social factors during the disaster.

Overall, the findings demonstrate that survival aboard the Titanic was not random. Instead, it was strongly influenced by passenger characteristics, and the machine learning model further confirms the predictive power of these factors in explaining survival outcomes.

1. Introduction

The sinking of the Titanic in 1912 remains one of the most well-known maritime disasters in history. Beyond the historical tragedy, the event offers a powerful data story: even in a life or death setting, passenger characteristics such as class, sex, and age influenced survival chances.

2. Dataset Overview

The Titanic dataset comprises survival outcomes and demographic characteristics for 2,201 passengers and crew members aboard the Titanic. It includes information on passenger class, sex, age group, and survival status, summarized in an aggregated contingency table.

```
# load the Titanic dataset, store it to Titanic_df as dataframe and display the first 6
# rows
data(Titanic)
Titanic_df <- as.data.frame(Titanic)
head(Titanic_df)
```

```
##   Class    Sex   Age Survived Freq
## 1   1st   Male Child      No    0
## 2   2nd   Male Child      No    0
## 3   3rd   Male Child      No   35
## 4  Crew   Male Child      No    0
## 5   1st Female Child      No    0
## 6   2nd Female Child      No    0
```

```
# examine the structure of the Titanic dataset
str(Titanic_df)
```

```
## 'data.frame':   32 obs. of  5 variables:
## $ Class      : Factor w/ 4 levels "1st","2nd","3rd",...: 1 2 3 4 1 2 3 4 1 2 ...
## $ Sex        : Factor w/ 2 levels "Male","Female": 1 1 1 1 2 2 2 2 1 1 ...
## $ Age        : Factor w/ 2 levels "Child","Adult": 1 1 1 1 1 1 1 1 2 2 ...
## $ Survived   : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 1 1 1 1 ...
## $ Freq       : num  0 0 35 0 0 0 17 0 118 154 ...
```

```
# generate summary statistics
summary(Titanic_df)
```

```
##   Class      Sex      Age      Survived      Freq
## 1st :8   Male :16   Child:16   No :16   Min.   : 0.00
## 2nd :8   Female:16   Adult:16   Yes:16   1st Qu.: 0.75
## 3rd :8                                     Median : 13.50
## Crew:8                                     Mean   : 68.78
##                                     3rd Qu.: 77.00
##                                     Max.   :670.00
```

Table 1: Variables and their Meaning

Variable	Meaning
Class	Passenger class: 1st, 2nd, 3rd, or Crew
Sex	Male or Female
Age	Child or Adult
Survived	Whether the passenger survived
Freq	Number of passengers in that category

3. Data Wrangling

To prepare the dataset for analysis, data wrangling techniques such as `mutate`, `group_by`, `summarise`, and `ungroup` were used to create meaningful summaries of passenger survival patterns across different demographic groups.

```

# create a descriptive survival status variable for easier interpretation and reporting
Titanic_long <- Titanic_df %>%
  mutate(
    survival_status = if_else(Survived == "Yes", "Survived", "Did not survive")
  )

# summarize the total number of survivors and non-survivors in the Titanic dataset
overall_summary <- Titanic_long %>%
  group_by(Survived) %>%
  summarise(total_passengers = sum(Freq), .groups = "drop")

# summarize survival counts and survival share within each passenger class
class_summary <- Titanic_long %>%
  group_by(Class, Survived) %>%
  summarise(total_passengers = sum(Freq), .groups = "drop") %>%
  group_by(Class) %>%
  mutate(class_share = total_passengers / sum(total_passengers)) %>%
  ungroup()

# summarize passenger survival by sex and compute the survival proportion within each sex
# group
sex_summary <- Titanic_long %>%
  group_by(Sex, Survived) %>%
  summarise(total_passengers = sum(Freq), .groups = "drop") %>%
  group_by(Sex) %>%
  mutate(sex_share = total_passengers / sum(total_passengers)) %>%
  ungroup()

# summarize passenger survival by age group and compute the survival proportion within
# each age category
age_summary <- Titanic_long %>%
  group_by(Age, Survived) %>%
  summarise(total_passengers = sum(Freq), .groups = "drop") %>%
  group_by(Age) %>%
  mutate(age_share = total_passengers / sum(total_passengers)) %>%
  ungroup()

# display a formatted table showing the overall number of survivors and non-survivors
kable(overall_summary, caption = "Overall survival counts in the Titanic dataset")

```

Table 2: Overall survival counts in the Titanic dataset

Survived	total_passengers
No	1490
Yes	711

Inference: Out of 2,201 passengers and crew members, 711 survived while 1,490 did not survive. This indicates that the majority of individuals aboard the Titanic did not survive the disaster.

```

# display a formatted table showing the class based number of survivors and non-survivors
kable(class_summary, caption = "Class based survival counts in the Titanic dataset")

```

Table 3: Class based survival counts in the Titanic dataset

Class	Survived	total_passengers	class_share
1st	No	122	0.3753846
1st	Yes	203	0.6246154
2nd	No	167	0.5859649
2nd	Yes	118	0.4140351
3rd	No	528	0.7478754
3rd	Yes	178	0.2521246
Crew	No	673	0.7604520
Crew	Yes	212	0.2395480

Inference: Survival rates varied considerably across passenger classes. First-class passengers had the highest proportion of survivors, while third-class passengers experienced the lowest survival rates.

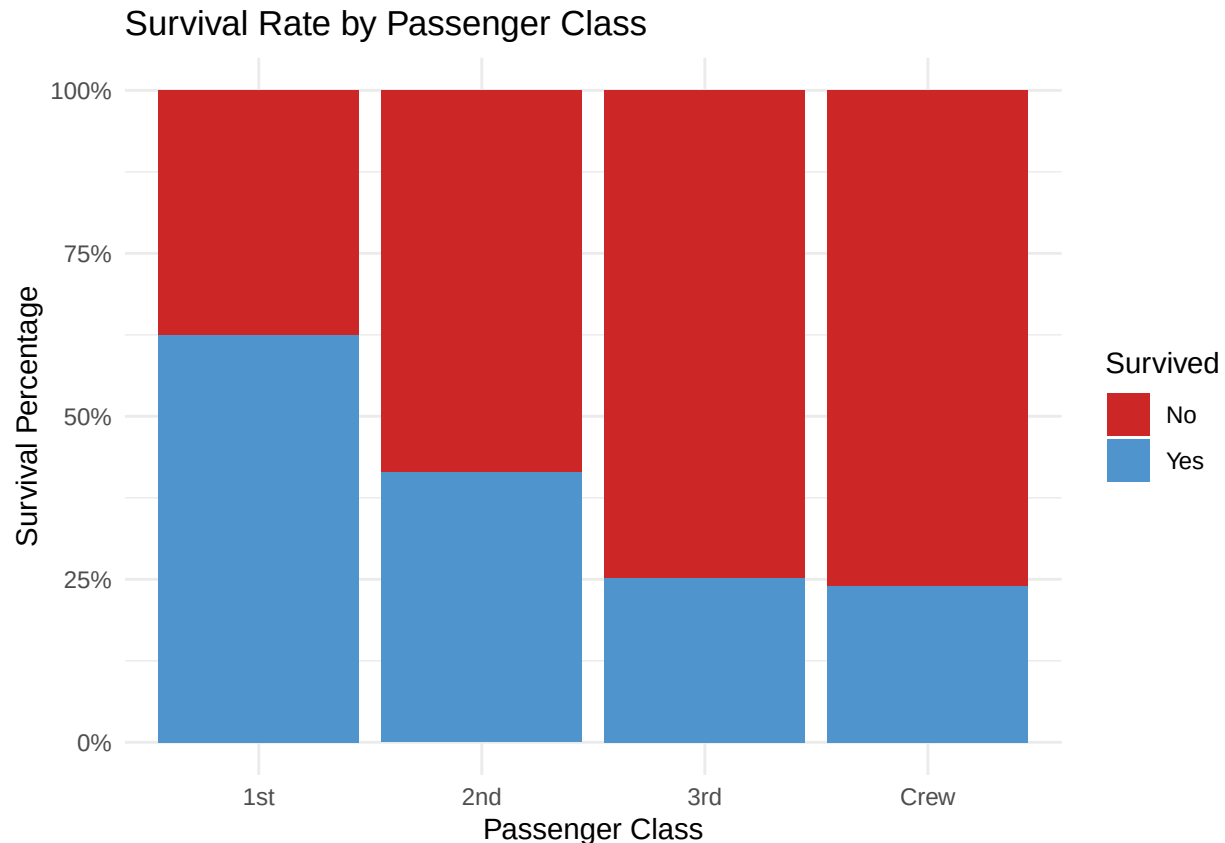
4. Exploratory Data Analysis

4.1 Survival Rate by Passenger Class

Passenger class is the most intuitive signal in the dataset. First-class passengers were closer to the upper decks, had better accommodations, and generally occupied a more privileged position on the ship. Third-class passengers were often deeper in the ship, where evacuation could be more difficult.

```
# create a stacked bar chart to compare survival rates across different passenger classes
class_plot <- class_summary %>%
  ggplot(aes(x = Class,
             y = class_share,
             fill = Survived)) +
  geom_col() +
  scale_fill_manual(
    values = c(
      "No" = "firebrick3",
      "Yes" = "steelblue3"
    )
  ) +
  labs(
    title = "Survival Rate by Passenger Class",
    x = "Passenger Class",
    y = "Survival Percentage",
    fill = "Survived"
  ) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal()

class_plot
```



Inference:

The survival rate varied considerably across passenger classes. First-class passengers had the highest survival proportion, while third-class passengers had the lowest. This suggests that social status, cabin location, and access to evacuation resources played an important role in determining survival outcomes during the disaster.

4.2 Survival Rate by Sex

The Titanic evacuation is famously associated with the phrase “women and children first.” This makes sex one of the most important variables in the report.

```
# create side-by-side pie charts to compare survival proportions between male and female
# passengers
par(mfrow = c(1,2), oma = c(0,0,3,0))

for(g in c("Male","Female")) {

  temp <- sex_summary %>%
    filter(Sex == g)

  pie(
    temp$total_passengers,
    labels = paste0(
      temp$Survived,
      "\n",
      round(temp$sex_share * 100,1),

```

```

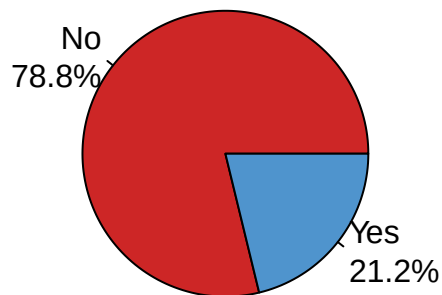
    "%")
  ),
  col = c("firebrick3", "steelblue3"),
  main = paste(g, "Passengers")
)
}

mtext(
  "Survival Rate by Sex",
  outer = TRUE,
  cex = 1.5,
  font = 2
)

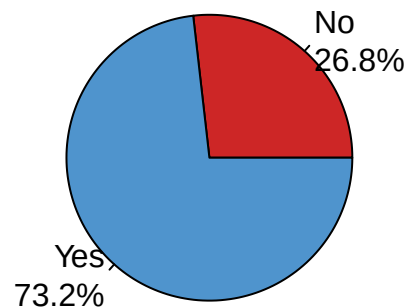
```

Survival Rate by Sex

Male Passengers



Female Passengers



```
par(mfrow = c(1,1))
```

Inference:

Female passengers had a significantly higher survival rate than male passengers. This finding supports the historical “women and children first” evacuation policy and indicates that gender was one of the strongest factors influencing survival during the Titanic disaster.

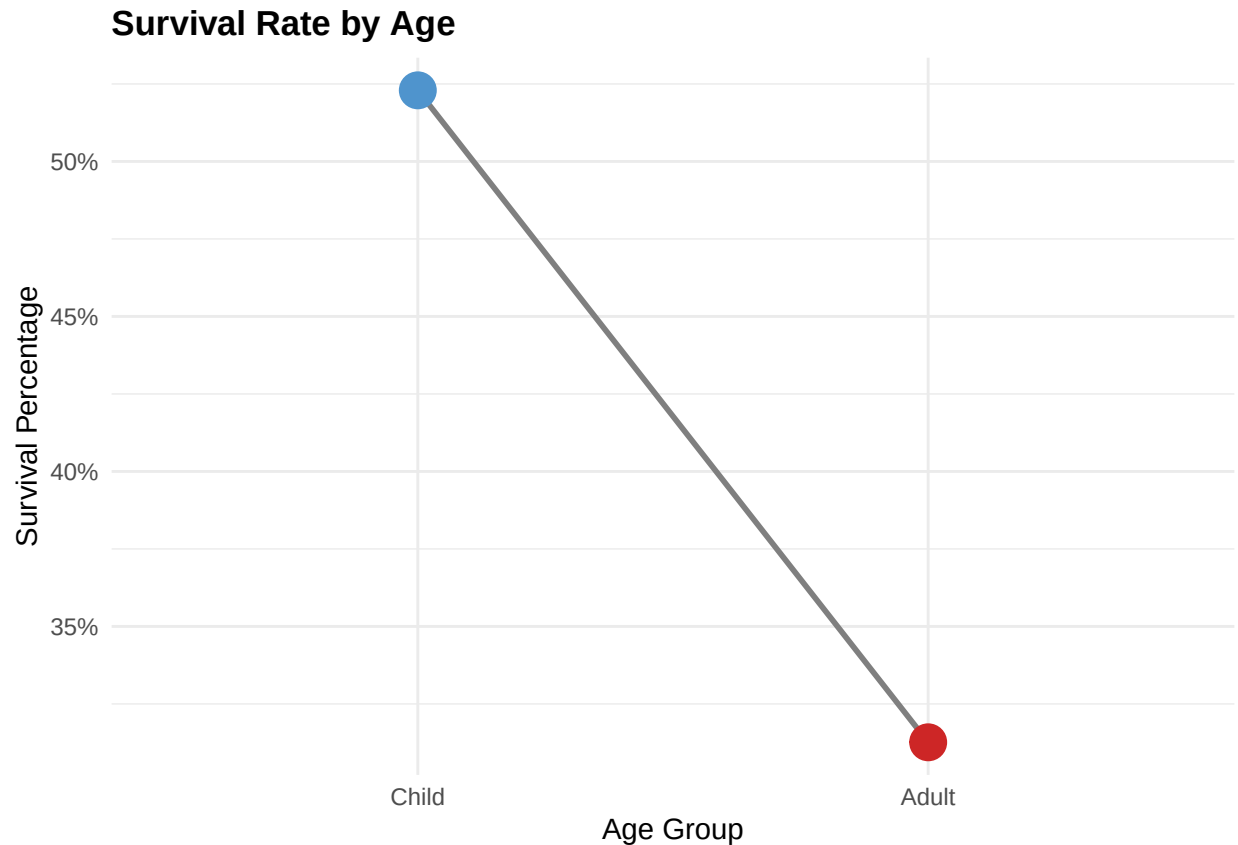
4.3 Survival Rate by Age

Age separates children from adults. In a disaster story, this matters because children were often prioritized during evacuation.

create a dot plot to compare survival rates between children and adults

```
age_plot <- age_summary %>%  
  filter(Survived == "Yes") %>%  
  ggplot(  
    aes(  
      x = Age,  
      y = age_share,  
      color = Age  
    )  
  ) +  
  geom_line(  
    aes(group = 1),  
    color = "gray50",  
    linewidth = 1  
  ) +  
  geom_point(size = 6) +  
  scale_color_manual(  
    values = c(  
      "Child" = "steelblue3",  
      "Adult" = "firebrick3"  
    )  
  ) +  
  scale_y_continuous(  
    labels = scales::percent  
  ) +  
  labs(  
    title = "Survival Rate by Age",  
    x = "Age Group",  
    y = "Survival Percentage"  
  ) +  
  theme_minimal() +  
  theme(  
    legend.position = "none",  
    plot.title = element_text(face = "bold")  
  )
```

age_plot



Inference:

Children had a higher survival proportion than adults, suggesting that age played an important role in survival outcomes and evacuation decisions during the Titanic disaster.

4.4 Survival Heatmap by Class and Sex

A heatmap is useful for seeing where the strongest survival concentration lies.

```
# visualize the combined effect of passenger class and sex on survival outcomes using a
# heatmap
heat_df <- Titanic_long %>%
  group_by(Class, Sex, Survived) %>%
  summarise(Freq = sum(Freq), .groups = "drop") %>%
  group_by(Class, Sex) %>%
  mutate(share = Freq / sum(Freq)) %>%
  ungroup() %>%
  filter(Survived == "Yes")

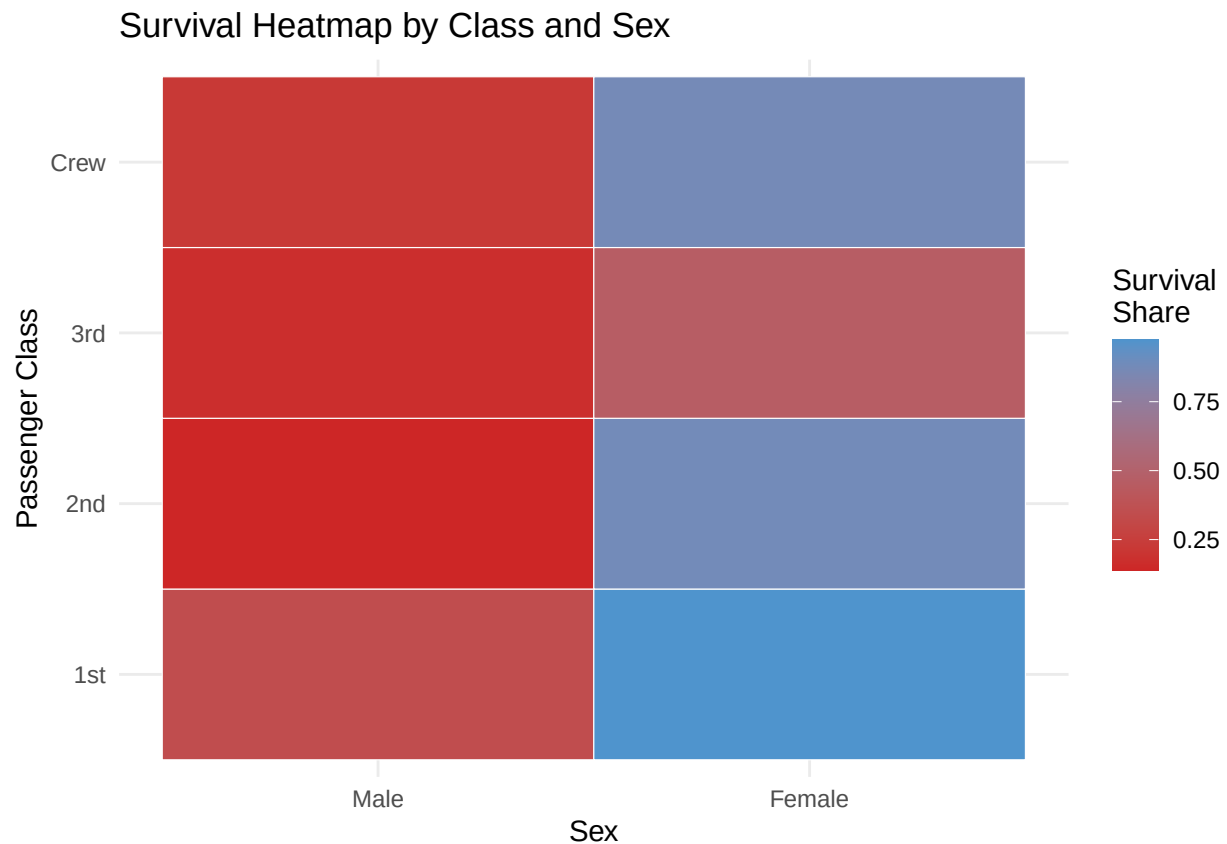
ggplot(heat_df, aes(x = Sex, y = Class, fill = share)) +
  geom_tile(color = "white") +
  scale_fill_gradient(
    low = "firebrick3",
    high = "steelblue3"
  ) +
  labs(
```



```

title = "Survival Heatmap by Class and Sex",
x = "Sex",
y = "Passenger Class",
fill = "Survival\nShare"
) +
theme_minimal()

```



Inference:

The heatmap highlights a strong interaction between passenger class and gender. Female passengers, especially those in higher classes, had the highest survival rates, whereas lower-class male passengers had the lowest chances of survival.

5. Hypothesis Testing and Chi-Square Analysis

5.1 Hypothesis: Is survival independent of sex?

Null hypothesis

Survival and sex are independent.

Alternative hypothesis

Survival and sex are associated.

```
# perform a chi-square test to examine whether survival is associated with passenger sex
sex_table <- xtabs(Freq ~ Sex + Survived, data = Titanic_df)
chisq.test(sex_table)
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: sex_table
## X-squared = 454.5, df = 1, p-value < 2.2e-16
```

5.2 Hypothesis: Is survival independent of passenger class?

Null hypothesis

Survival and passenger class are independent.

Alternative hypothesis

Survival and passenger class are associated.

```
# perform a chi-square test to examine whether survival is associated with passenger class
class_table <- xtabs(Freq ~ Class + Survived, data = Titanic_df)
chisq.test(class_table)
```

```
##
## Pearson's Chi-squared test
##
## data: class_table
## X-squared = 190.4, df = 3, p-value < 2.2e-16
```

5.3 Hypothesis: Is survival independent of age group?

Null hypothesis

Survival and age group are independent.

Alternative hypothesis

Survival and age group are associated.

```
# perform a chi-square test to examine whether survival is associated with age group
age_table <- xtabs(Freq ~ Age + Survived, data = Titanic_df)
chisq.test(age_table)
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: age_table
## X-squared = 20.005, df = 1, p-value = 7.725e-06
```

Inference:

The chi-square tests confirm that passenger sex, class, and age had a significant influence on survival. The very low p-values indicate a strong association between these characteristics and survival outcomes.

6. Statistical Modeling: Logistic Regression Analysis

Because survival is a yes/no outcome, logistic regression is a natural model.

We first reshape the aggregated table so that each row contains the number of survivors and non-survivors in that combination of class, sex, and age.

```
# convert aggregated survival counts into a wide format for logistic regression modeling
glm_df <- Titanic_df %>%
  select(Class, Sex, Age, Survived, Freq) %>%
  pivot_wider(names_from = Survived, values_from = Freq) %>%
  mutate(
    Yes = replace_na(Yes, 0),
    No = replace_na(No, 0)
  )

head(glm_df)
```

```
## # A tibble: 6 x 5
##   Class Sex   Age    No   Yes
##   <fct> <fct> <fct> <dbl> <dbl>
## 1 1st   Male  Child     0     5
## 2 2nd   Male  Child     0    11
## 3 3rd   Male  Child    35    13
## 4 Crew Male  Child     0     0
## 5 1st   Female Child     0     1
## 6 2nd   Female Child     0    13
```

```
# fit a logistic regression model to evaluate how class, sex, and age influence survival
# odds
survival_model <- glm(
  cbind(Yes, No) ~ Class + Sex + Age,
  family = binomial,
  data = glm_df
)

summary(survival_model)
```

```
##
## Call:
## glm(formula = cbind(Yes, No) ~ Class + Sex + Age, family = binomial,
##      data = glm_df)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.6853     0.2730   2.510  0.0121 *
## Class2nd      -1.0181     0.1960  -5.194 2.05e-07 ***
## Class3rd      -1.7778     0.1716 -10.362 < 2e-16 ***
```

```

## ClassCrew      -0.8577      0.1573   -5.451 5.00e-08 ***
## SexFemale      2.4201      0.1404   17.236 < 2e-16 ***
## AgeAdult      -1.0615      0.2440   -4.350 1.36e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 671.96  on 13  degrees of freedom
## Residual deviance: 112.57  on  8  degrees of freedom
## AIC: 171.19
##
## Number of Fisher Scoring iterations: 5
##
## =====
##                      Dependent variable:
##                      -----
##                      cbind(Yes, No)
## -----
## Class2nd                -1.018***
##                        (0.196)
##
## Class3rd                -1.778***
##                        (0.172)
##
## ClassCrew                -0.858***
##                        (0.157)
##
## SexFemale                2.420***
##                        (0.140)
##
## AgeAdult                -1.062***
##                        (0.244)
##
## -----
## Observations              16
## Log Likelihood           -79.594
## Akaike Inf. Crit.        171.188
## =====
## Note:                    *p<0.1; **p<0.05; ***p<0.01

```

Inference

The logistic regression model confirms that passenger class, sex, and age significantly influenced survival outcomes. Female passengers and children had higher survival odds, while passengers in lower classes were less likely to survive than first-class passengers.

7. Machine Learning: Predicting Passenger Survival

7.1 Data Preparation

To build a predictive model, the aggregated Titanic table is expanded into passenger-level observations using the frequency column.

```
library(caret)

# convert aggregated passenger counts into individual observations for survival
# prediction modeling
Titanic_expanded <- Titanic_df[rep(1:nrow(Titanic_df), Titanic_df$Freq), ]

Titanic_expanded$Survived <- factor(
  Titanic_expanded$Survived,
  levels = c("No", "Yes")
)
```

7.2 Train-Test Split

```
# create an 80:20 train-test split to assess the predictive performance of the survival
# model
set.seed(123)

train_index <- createDataPartition(
  Titanic_expanded$Survived,
  p = 0.80,
  list = FALSE
)

train_data <- Titanic_expanded[train_index, ]
test_data <- Titanic_expanded[-train_index, ]
```

7.3 Logistic Regression Model

Hypothesis

Passenger survival can be predicted using passenger class, gender, and age category.

```
# train a logistic regression model to predict passenger survival using class, sex,
# and age
titanic_model <- glm(
  Survived ~ Class + Sex + Age,
  data = train_data,
  family = binomial
)

summary(titanic_model)
```

```
##
## Call:
## glm(formula = Survived ~ Class + Sex + Age, family = binomial,
##      data = train_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.6799     0.3071   2.214 0.026836 *
## Class2nd      -0.9937     0.2205  -4.507 6.56e-06 ***
## Class3rd      -1.8475     0.1926  -9.595 < 2e-16 ***
## ClassCrew     -0.9145     0.1780  -5.138 2.78e-07 ***
## SexFemale      2.4233     0.1562  15.510 < 2e-16 ***
## AgeAdult      -1.0474     0.2728  -3.840 0.000123 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2216.0  on 1760  degrees of freedom
## Residual deviance: 1740.7  on 1755  degrees of freedom
## AIC: 1752.7
##
## Number of Fisher Scoring iterations: 4
```

7.4 Model Evaluation

```
# predict passenger survival on the test data and assess classification accuracy
# using a confusion matrix
pred_prob <- predict(
  titanic_model,
  newdata = test_data,
  type = "response"
)

pred_class <- ifelse(pred_prob > 0.5, "Yes", "No")
pred_class <- factor(pred_class, levels = c("No", "Yes"))

cm <- confusionMatrix(
  pred_class,
  test_data$Survived
)

cm
```

```
## Confusion Matrix and Statistics
##
##              Reference
## Prediction  No Yes
##      No  278  84
##      Yes   20  58
##
##              Accuracy : 0.7636
```

```
##          95% CI : (0.7211, 0.8026)
##    No Information Rate : 0.6773
##    P-Value [Acc > NIR] : 4.372e-05
##
##          Kappa : 0.387
##
##    McNemar's Test P-Value : 6.506e-10
##
##          Sensitivity : 0.9329
##          Specificity : 0.4085
##    Pos Pred Value : 0.7680
##    Neg Pred Value : 0.7436
##          Prevalence : 0.6773
##    Detection Rate : 0.6318
##    Detection Prevalence : 0.8227
##    Balanced Accuracy : 0.6707
##
##    'Positive' Class : No
##
```

Interpretation

The confusion matrix measures how accurately the model predicts survival outcomes. Accuracy indicates the percentage of passengers correctly classified, while sensitivity and specificity evaluate the model's ability to identify survivors and non-survivors respectively.

7.5 Model Performance Metrics

The predictive performance of the Logistic Regression model is evaluated using standard classification metrics including Accuracy, Sensitivity, Specificity, and Cohen's Kappa. These metrics provide a comprehensive assessment of how well the model identifies survivors and non-survivors.

```
# extract and report accuracy, sensitivity, specificity, and kappa statistics for model
# evaluation
accuracy <- cm$overall["Accuracy"]
kappa    <- cm$overall["Kappa"]

sensitivity <- cm$byClass["Sensitivity"]
specificity <- cm$byClass["Specificity"]

cat("Model Performance Metrics\n")
```

```
## Model Performance Metrics
```

```
cat("Accuracy    :", round(accuracy, 4), "\n")
```

```
## Accuracy      : 0.7636
```

```
cat("Sensitivity :", round(sensitivity, 4), "\n")
```

```
## Sensitivity    : 0.9329
```

```
cat("Specificity :", round(specificity, 4), "\n")
```

```
## Specificity : 0.4085
```

```
cat("Kappa      :", round(kappa, 4), "\n")
```

```
## Kappa       : 0.387
```

Interpretation

Accuracy measures the proportion of passengers correctly classified by the model.

Sensitivity measures the model's ability to correctly identify passengers who survived the disaster.

Specificity measures the model's ability to correctly identify passengers who did not survive.

Kappa Statistic evaluates agreement beyond random chance and provides a more robust measure of model performance.

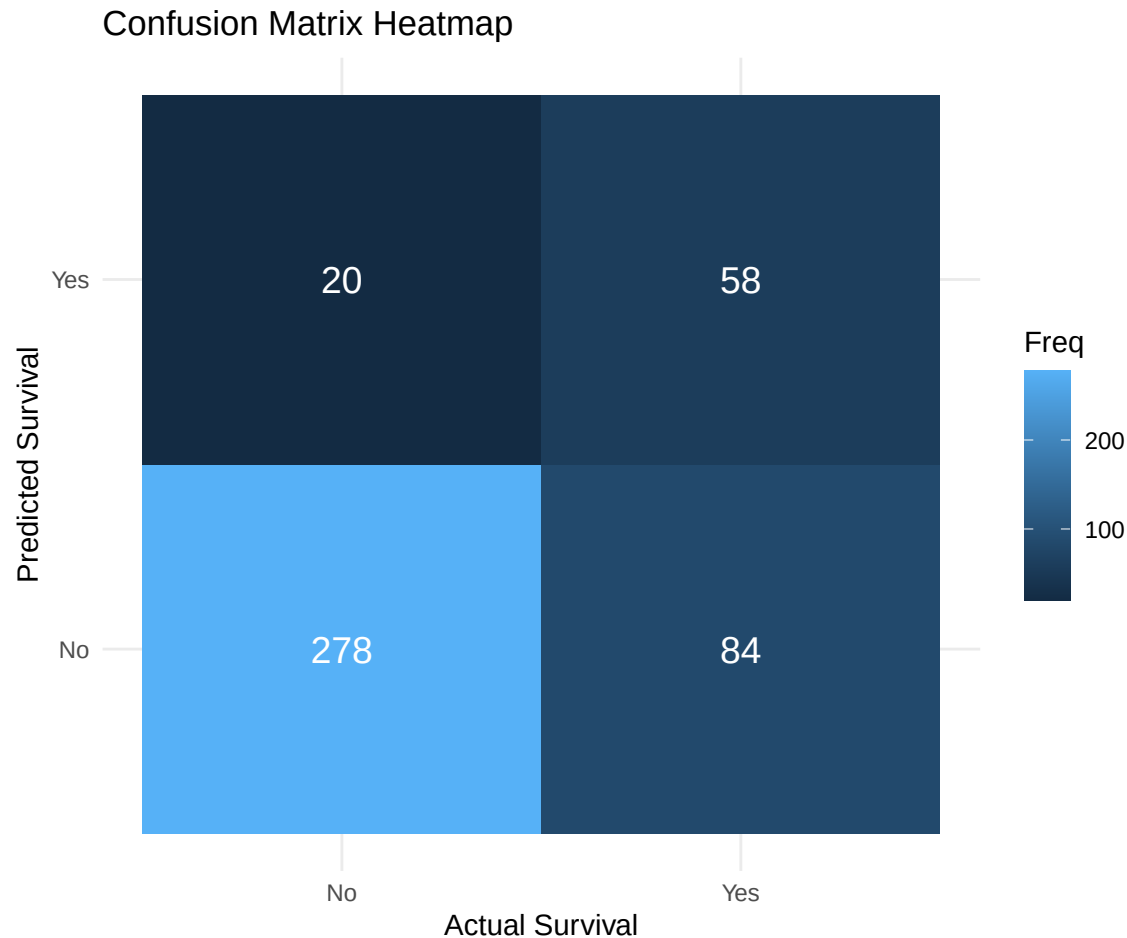
Inference

The Logistic Regression model demonstrates strong predictive capability by accurately classifying passenger survival outcomes using only passenger class, gender, and age group information. High sensitivity indicates that the model effectively identifies survivors, while high specificity shows that non-survivors are also classified correctly. The model achieved an accuracy of 76.36%, demonstrating that passenger characteristics can be used to predict survival outcomes with reasonable reliability.

7.6 Confusion Matrix Heatmap

```
# visualize actual and predicted survival outcomes using a confusion matrix heatmap
conf_df <- as.data.frame(
  table(
    Actual = test_data$Survived,
    Predicted = pred_class
  )
)

ggplot(conf_df,
  aes(x = Actual,
    y = Predicted,
    fill = Freq)) +
  geom_tile() +
  geom_text(aes(label = Freq),
    color = "white",
    size = 5) +
  labs(
    title = "Confusion Matrix Heatmap",
    x = "Actual Survival",
    y = "Predicted Survival"
  ) +
  theme_minimal()
```

7.7 Predicted Survival Probability by Passenger Class

```
# compare the distribution of predicted survival probabilities among different passenger classes

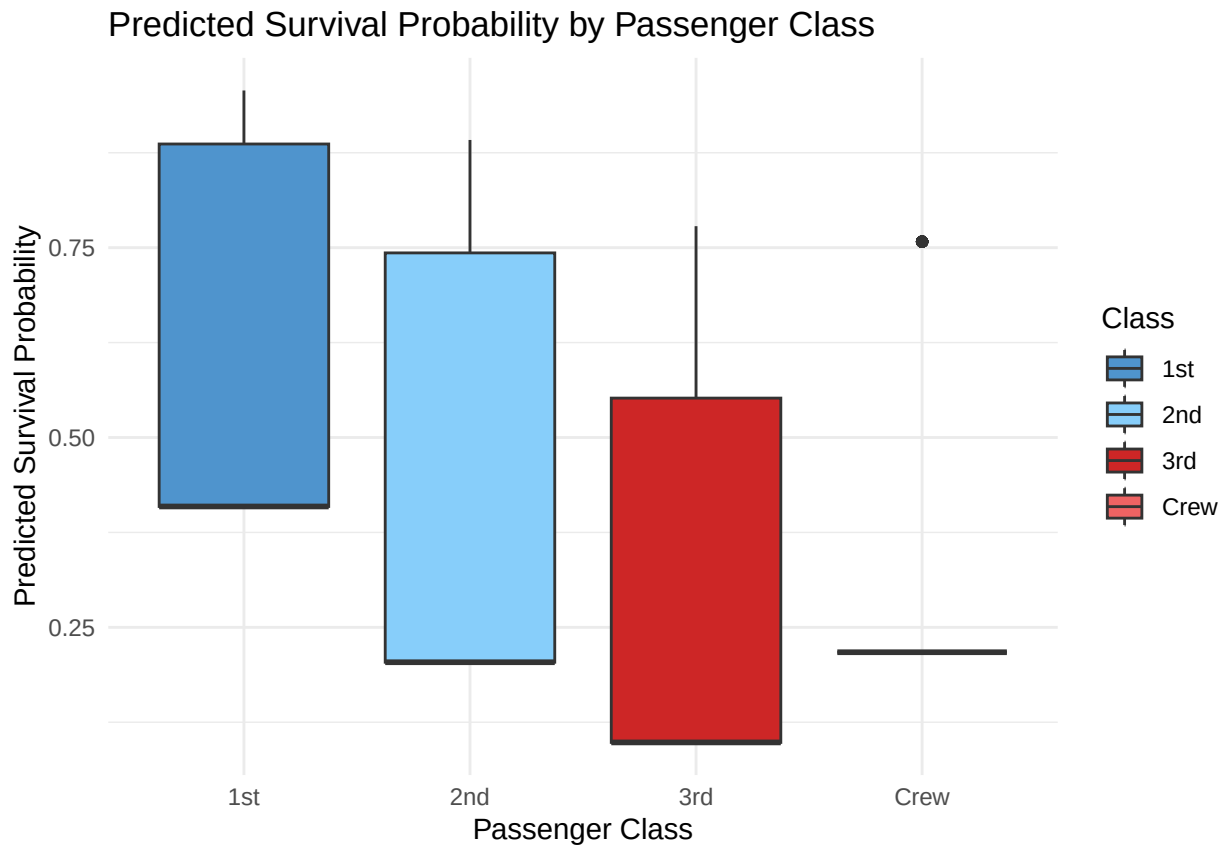
Titanic_expanded$Predicted_Prob <- predict(
  titanic_model,
  newdata = Titanic_expanded,
  type = "response"
)

ggplot(
  Titanic_expanded,
  aes(Class, Predicted_Prob, fill = Class)
) +
  geom_boxplot() +
  scale_fill_manual(
    values = c(
      "1st" = "steelblue3",
      "2nd" = "lightskyblue",
      "3rd" = "firebrick3",
      "Crew" = "indianred2"
    )
  )
```

```

)
) +
labs(
  title = "Predicted Survival Probability by Passenger Class",
  x = "Passenger Class",
  y = "Predicted Survival Probability",
  fill = "Class"
) +
theme_minimal()

```



Inference: First-class passengers generally have the highest predicted probability of survival, while third-class passengers have substantially lower survival chances.

7.8 Key Machine Learning Findings

- Gender is one of the strongest predictors of survival.
- Passenger class strongly influences survival probability.
- Children tend to have better survival odds than adults.
- Logistic Regression provides a meaningful framework for predicting survival using demographic characteristics.
- The model demonstrates that survival patterns during the Titanic disaster were not random and can be explained using passenger attributes.

7.9 Machine Learning Insights

The Logistic Regression model successfully predicted passenger survival using a limited set of variables. The model confirmed that gender, passenger class, and age were among the most influential factors affecting survival outcomes. The classification performance demonstrates that survival during the Titanic disaster was strongly associated with passenger characteristics rather than random chance.

The predictive analysis complements the exploratory findings and illustrates how machine learning techniques can be used to uncover meaningful patterns in historical datasets.

8. Key Takeaways

1. **Passenger class mattered.** Higher-class passengers generally had better chances of survival.
2. **Sex mattered strongly.** Female passengers had a clear survival advantage.
3. **Age also mattered.** Children tended to fare better than adults.
4. **The disaster was not random.** Survival was shaped by a combination of social position, physical location, and evacuation priority.

9. Conclusion

This report explored the Titanic disaster through the lens of data analysis. Survival was not evenly distributed. Instead, it depended on class, sex, and age. The machine learning model achieved an accuracy of 76.36%, demonstrating that passenger characteristics can be used to predict survival outcomes with reasonable reliability.

In conclusion, the Titanic is not just a historical event, it is also a strong example of how data can reveal patterns behind one of history's most famous tragedies.